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Third Semester B.E./B.Tech. Degree Examination, June/July 2025

Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

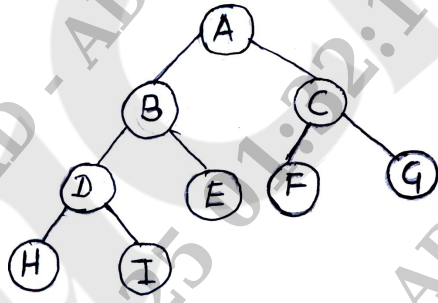
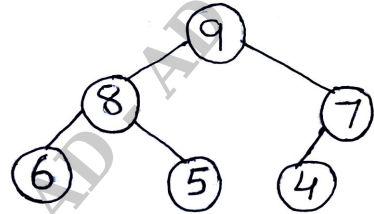
Module – 1			M	L	C
Q.1	a.	Define data structure. With a neat diagram, explain the classification of data structure.	5	L1	CO1
	b.	Explain dynamic memory allocation functions with suitable examples.	5	L2	CO1
	c.	For the given sparse matrix draw the triplet representation and also draw the transpose of resultant triplet. $A = \begin{bmatrix} 15 & 0 & 0 & 22 & 0 & -15 \\ 0 & 11 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 91 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 28 & 0 & 0 & 0 \end{bmatrix}$	10	L3	CO1
OR					
Q.2	a.	Define stack. With suitable example write the array representation of stack.	5	L1	CO2
	b.	Write a C functions to implement push(), pop() and display() operations for stack using array.	5	L2	CO2
	c.	Translate the following infix expressions to postfix form using stack: i) $((A * B) + C) / D$ ii) $A * B * C + D$	10	L3	CO2
Module – 2					
Q.3	a.	What are the disadvantages of linear queue?	5	L1	CO2
	b.	With suitable example discuss the representation of linear queue with array.	5	L2	CO2
	c.	Develop functions in C to implement insertion, deletion and display operations on circular queue of integers.	10	L3	CO2
OR					
Q.4	a.	What is linked list? With suitable examples explain different types of linked lists.	5	L1	CO3
	b.	Write a C functions to implement a stack of integers using a Singly Linked List (SLL).	5	L2	CO3

c.	Develop a functions in C for the following operations on singly linked list of integers: i) Insert a element at end of SLL ii) Delete a element at end of SLL iii) Concatenation of two SLL	10	L3	CO3
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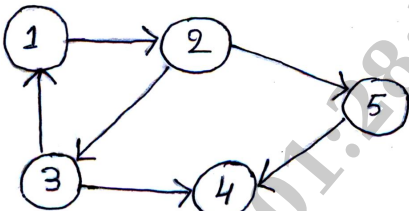
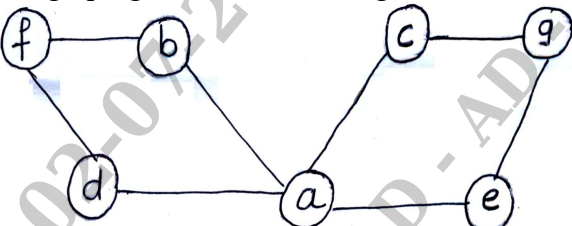
Module – 3

Q.5	a.	Write a structure definition for Doubly Linked List (DLL) of integers. What are the advantages of DLL over SLL?	4	L1	CO3
	b.	Develop a C functions for the following operations on DLL of integers: i) Insert a node at front of DLL ii) Delete a node at end of DLL	10	L3	CO3
	c.	For the given sparse matrix design the linked list representation. $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 1 \\ 0 & 0 & 6 & 0 \end{bmatrix}$	6	L4	CO3

OR

Q.6	a.	Define binary tree. Write array and linked list representation for given binary tree.  Fig.Q.6(a)	4	L1	CO4
	b.	Develop recursive C functions for inorder, preorder and postorder traversal of a binary tree. Find inorder, preorder and postorder traversals for the given binary tree.  Fig.Q.6 (b)	10	L3	CO4
	c.	Design threaded binary tree for the given elements 10, 20, 30, 40, 50	6	L4	CO4

Module – 4

Q.7	a.	Write an adjacency matrix and adjacency linked list representation for following given graph.	6	L1	CO4
 Fig.Q.7(a)					
	b.	Develop a C function to traverse a graph using Depth First Search (DFS). Apply DFS for the graph given below starting from C.	8	L4	CO4
 Fig.Q.7(b)					
	c.	List and explain different types of selection trees with suitable examples.	6	L4	CO4
OR					
Q.8	a.	Define forest data structure. With a suitable example write a procedure to transform forest into binary tree.	6	L1	CO4
	b.	For a given data design a binary search tree. Apply inorder, preorder and postorder traversals on resultant binary search tree. 6, 9, 5, 2, 8, 15, 24, 14, 7, 8, 5, 2	8	L4	CO4
	c.	Develop a C functions to perform the following operations on Binary Search Tree (BST): i) Inserting an element into BST ii) Recursive search of given key element on BST.	6	L4	CO4
Module – 5					
Q.9	a.	Explain hashing with suitable example. Explain different types of hashing functions in details.	10	L2	CO5
	b.	Explain static hashing and dynamic hashing in detail.	10	L2	CO5
OR					
Q.10		Write a note on:			
	a.	Leftist trees	6	L2	CO5
	b.	Optimal binary search trees	7	L2	CO5
	c.	Priority queues.	7	L2	CO5
